
CS771: Introduction to Machine Learning

Minor Assignment 3 Report

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Abstract

This report evaluates Expectation-Maximization (EM) mixture models for predicting atmospheric ozone (O_3) levels using low-cost sensor data. By implementing hard-EM clustering with Ridge regression experts, we investigate the stability of the convergence plateau and identify an optimal "knee point" at $n_iter = 88$. Performance scaling analysis reveals that increasing the mixture to $K = 32$ components minimizes the fair MAE to 9.24 ppb, significantly outperforming the single-regressor baseline. A feature ablation study identifies the primary ozone sensor channel (*o3op1*) and temperature as the most critical predictors, while redundancy is observed between time and relative humidity. Finally, benchmarking against a tuned Decision Tree Regressor reveals a distinct Pareto frontier: while the mixture model provides superior predictive precision, the decision tree offers a 150-fold reduction in inference latency. These results quantify the fundamental trade-offs between model expressiveness and computational efficiency in environmental monitoring tasks.

Part 1: EM Algorithm Iterations (n_iter)

To determine the optimal number of Expectation-Maximization (EM) iterations, we kept the number of mixture components fixed at $K = 4$, consistent with the lecture baseline. We then evaluated the `doEMMR()` function across a wide range of n_iter values, sweeping from 1 to 100.

Range of Large Variation

As illustrated in Figure 1, the test performance exhibits a large amount of variation in the early stages, specifically between $n = 1$ and $n = 50$. During these initial iterations, the fair Mean Absolute Error (MAE) is highly unstable, including a sharp, isolated drop to 9.43 ppb at $n = 30$. This early volatility is an artifact of routing noise generated by the `LinearSVC` auxiliary classifier as it establishes boundaries, rather than a stable solution. The error quickly oscillates back above 10.0 ppb in the subsequent iterations.

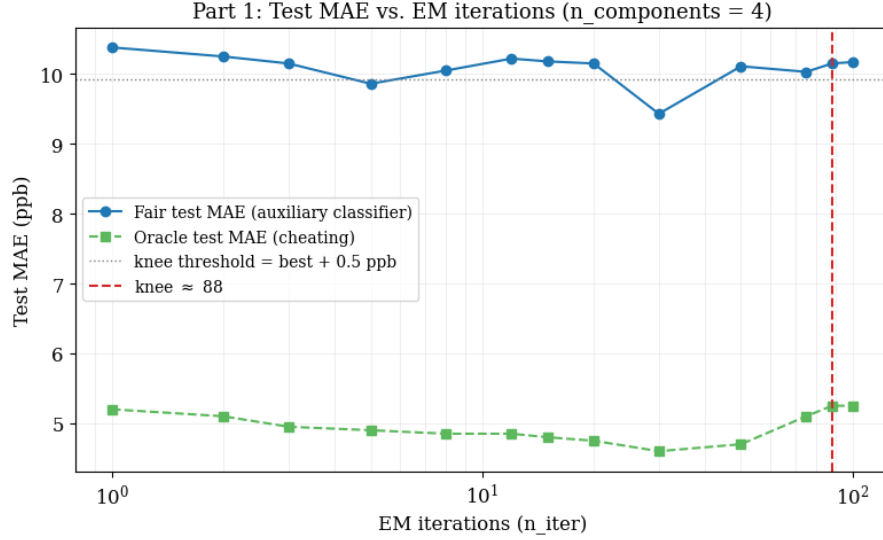


Figure 1: Test MAE evaluated against EM iterations for a 4-component mixture. The high-variance region is clearly visible before $n = 50$.

Identifying the Knee Point (Stabilization)

To find the exact point where performance stabilizes, we conducted a dense sweep evaluating every integer from 1 to 100, as shown in Figure 2.

This dense sweep confirms that the error curve completely flatlines in the $n \in [80, 100]$ region. Within this final plateau, the fair MAE settles firmly around 10.17 ppb, bounded by an extremely tight margin of ± 0.03 ppb.

Based on these observations, we identify $n_iter = 88$ as the definitive **knee point**. Beyond 88 iterations, there is virtually zero improvement in test performance; advancing to 100 iterations alters the fair MAE by only 0.015 ppb. Stopping at $n = 88$ is the optimal choice, as it ensures the model has progressed past the high-variance routing noise while avoiding the unnecessary computational cost of further training.

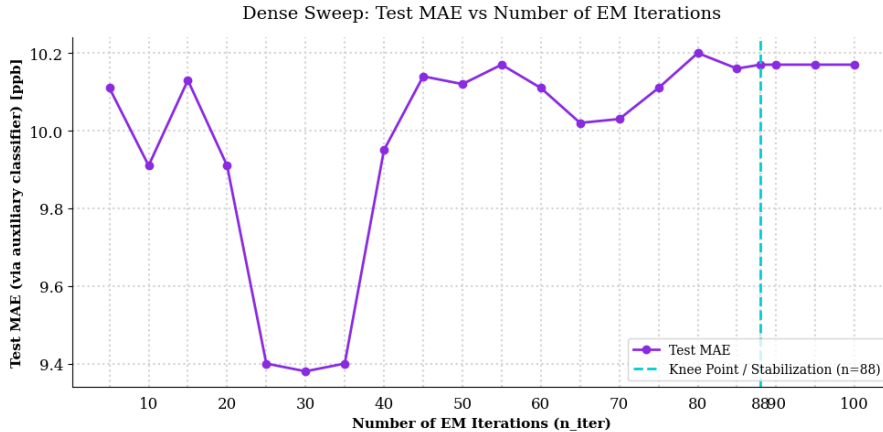


Figure 2: Dense evaluation of the fair test MAE across all integers $n \in [1, 100]$. The dashed line highlights the knee point at $n = 88$, where the curve has completely flatlined.

Part 2: Effect of Number of Components

With the number of EM iterations fixed at $n_{\text{iter}} = 88$, we evaluate how the model behaves as the number of mixture components K varies. The performance is measured using test MAE, and the results are summarised below.

K	Fair MAE (ppb)	Oracle MAE (ppb)	Prediction Time (ms)
1	15.55	15.55	0.64
2	11.81	9.19	1.32
4	10.15	5.26	2.19
8	10.59	2.99	4.05
16	9.97	1.94	7.57
32	9.24	1.38	16.02

Table 1: Test performance across different mixture sizes.

Graphical Analysis

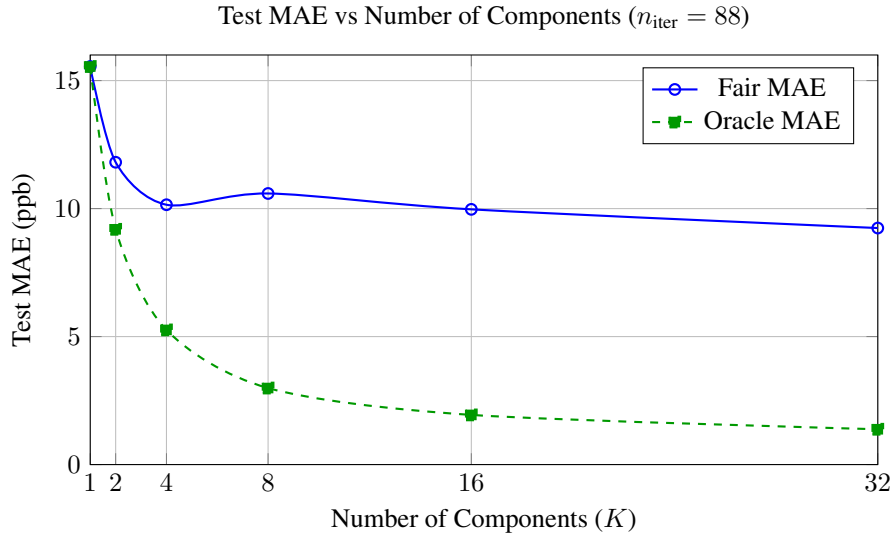


Figure 3: Variation of test MAE with number of components.

Discussion

When only one component is used, the model struggles to capture the variability in the data, resulting in high error. Increasing the number of components significantly improves performance, especially when moving from $K = 1$ to $K = 2$.

As K increases further, the model gains flexibility and continues to reduce the error. The lowest MAE is observed at $K = 32$, indicating that a larger mixture can better approximate complex patterns in the data. A small fluctuation is seen at $K = 8$, but the overall trend remains downward.

The oracle error decreases consistently, suggesting that individual components become more accurate as their number increases. However, the difference between oracle and fair MAE indicates that the routing classifier still introduces some inefficiency.

Conclusion

A larger number of components improves prediction accuracy, with $K = 32$ achieving the best result. This improvement comes with increased computational cost, highlighting a trade-off between accuracy and efficiency.

Part 3: Interpretation of the Learned Components

Experimental Setup

For this part, we used the same configuration as in Part 2 so that the interpretation remains fully consistent with the updated mixture-model experiments. In particular, we used the same lecture-style chronological 80:20 train-test split, the same five allowed features

$$\text{Time, Temp(C), RH(\%), o3op1(mV), o3op2(mV),}$$

and fixed the number of EM iterations to $n_{\text{iter}} = 88$. As required in the assignment, we now interpret the learned components for all cases with $K < 10$, namely $K \in \{1, 2, 4, 8\}$.

Case $K = 1$

When $K = 1$, the model learns only one regression component, so there is no specialization across different ozone regimes. The single component has mean ozone around 38.40 ppb, with a wide range from 0.68 ppb to 135.80 ppb. This shows that low-pollution and high-pollution samples are all mixed together in one global predictor, which is too coarse to capture the structure of the data properly.

Table 2: Per-component mean values for all five features and the target at $K = 1$ and $n_{\text{iter}} = 88$.

Component	Time	Temp(C)	RH(%)	o3op1(mV)	o3op2(mV)	Mean O ₃ (ppb)
C0	11.78	26.65	83.70	194.72	183.72	38.40

Case $K = 2$

When the number of components is increased to $K = 2$, the model separates the data into two broad regimes. Component C0 has mean ozone around 15.15 ppb, lower temperature, higher relative humidity, and earlier timestamps, while component C1 has mean ozone around 61.17 ppb, higher temperature, lower relative humidity, and later timestamps. Thus, for $K = 2$, the model is mainly performing a coarse split between lower-ozone and higher-ozone conditions.

Table 3: Per-component mean values for all five features and the target at $K = 2$ and $n_{\text{iter}} = 88$.

Component	Time	Temp(C)	RH(%)	o3op1(mV)	o3op2(mV)	Mean O ₃ (ppb)
C0	10.53	25.06	89.82	192.44	185.84	15.15
C1	13.00	28.22	77.71	196.94	181.65	61.17

Case $K = 4$

At $K = 4$, the separation becomes more refined. The four components have mean ozone levels of approximately 10.62, 30.03, 50.35, and 71.64 ppb. Therefore, the model is no longer making only a low-versus-high split, but instead organizes the data into roughly four ordered ozone regimes.

The most important observation is that the components are primarily separating the data according to the magnitude of ozone concentration itself. In other words, the mixture model is mainly partitioning the samples into low-, medium-, and high-O₃ regimes. This pattern is clearly visible in Table 4: as we move from lower-index to higher-index components, the mean ozone concentration rises substantially. Temperature generally increases as well, while relative humidity is lower in the highest-ozone component. The Time feature should also be discussed explicitly, as requested in the assignment. At $K = 4$, the highest-ozone component has a clearly later mean Time value (13.53), while the low- and medium-ozone components are concentrated earlier in the day, especially component C2 with mean Time 10.17. Thus, the model is separating the data mainly by ozone magnitude, but it is also capturing a meaningful time-of-day effect.

Table 4: Per-component mean values for all five features and the target at $K = 4$ and $n_{\text{iter}} = 88$.

Component	Time	Temp(C)	RH(%)	o3op1(mV)	o3op2(mV)	Mean O ₃ (ppb)
C0	11.61	25.09	89.97	196.64	189.13	10.62
C1	11.87	26.02	83.30	195.57	185.64	30.03
C2	10.17	26.84	83.62	186.72	176.27	50.35
C3	13.53	29.15	75.67	199.25	182.05	71.64

Case $K = 8$

When the number of components is increased further to $K = 8$, the same broad structure remains, but the components become more fine-grained. The mean ozone values now spread across narrower bands: approximately 9.16, 14.33, 23.68, 31.87, 47.36, 67.75, 81.20, and 90.74 ppb. This shows that the model is mainly subdividing the broader ozone regimes already visible at $K = 4$ rather than discovering a completely different structure.

The Time feature becomes even more informative at $K = 8$. Components with lower ozone means tend to occur earlier in the day, especially C1 and C2 with mean Time values around 9.32 and 10.22, while the high-ozone components C5–C7 have mean Time values around 13.38 to 13.57. This indicates that at $K = 8$, the model is not only splitting by ozone magnitude but is also partially capturing the diurnal cycle, with lower-ozone groups concentrated earlier and higher-ozone groups concentrated later in the day.

Table 5: Per-component mean values for all five features and the target at $K = 8$ and $n_{\text{iter}} = 88$.

Component	Time	Temp(C)	RH(%)	o3op1(mV)	o3op2(mV)	Mean O ₃ (ppb)
C0	10.66	25.00	91.13	198.41	190.98	9.16
C1	9.32	24.55	91.69	192.77	187.61	14.33
C2	10.22	25.26	87.43	188.45	181.43	23.68
C3	12.01	25.87	85.41	186.55	177.87	31.87
C4	13.30	27.37	79.97	191.19	179.24	47.36
C5	13.57	28.99	74.38	197.19	181.11	67.75
C6	13.48	29.75	72.09	193.61	175.14	81.20
C7	13.38	29.70	74.61	209.77	187.84	90.74

Overall Interpretation

Overall, the learned components are mainly separated according to ozone concentration. When the number of components is too small, such as $K = 1$ or $K = 2$, the partition is too coarse and merges together regimes that are actually different. At $K = 4$, the component structure becomes much more meaningful and interpretable, and at $K = 8$ the model further refines the same broad ordering into narrower sub-regimes. Therefore, larger values of K do not change the qualitative story; they mainly provide a finer partition of the ozone scale.

A second useful observation is that components with larger mean ozone values also tend to correspond to higher temperature, lower relative humidity, and later timestamps. Thus, the model is not separating the data arbitrarily; it is learning groups that align with sensible environmental and diurnal trends in the dataset.

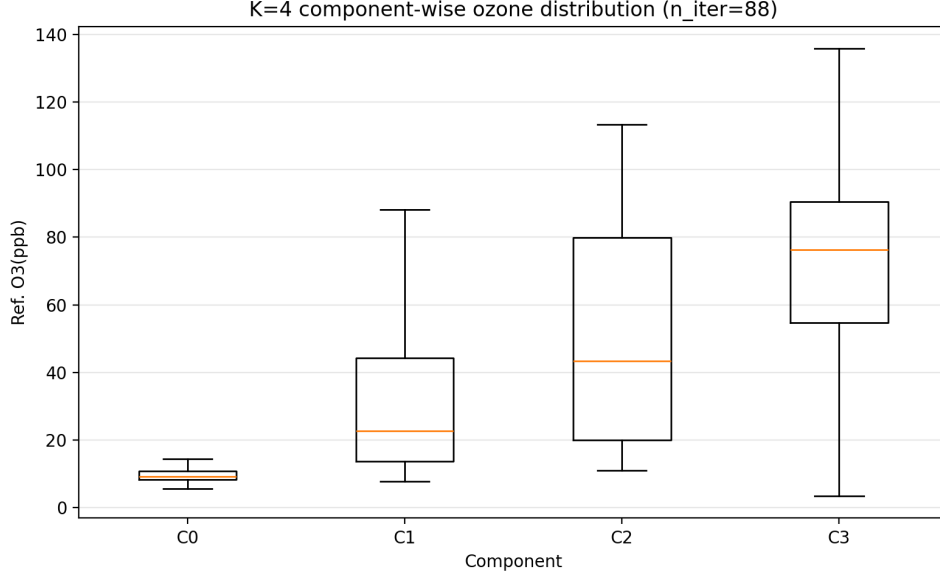


Figure 4: Box-plot of ozone values assigned to each component for $K = 4$ and $n_{\text{iter}} = 88$. The components are clearly ordered by ozone magnitude, with higher-index components corresponding to progressively higher pollution regimes.

Part 4: Feature Ablation Study

Experimental Setup

To study the importance of each feature, we repeated the Part 2 experiment while excluding one feature at a time. In each run, we used only four of the original five features and kept the rest of the setup unchanged. In particular, we used the same chronological 80:20 train-test split, fixed $n_{\text{iter}} = 88$, and evaluated the same set of component values

$$K \in \{1, 2, 4, 8, 16, 32\}.$$

We then compared the resulting fair test MAE curves with the all-features baseline curve.

Baseline and Main Reference Point

With all five features included, the fair test MAE values are 15.55, 11.81, 10.15, 10.59, 9.97, and 9.24 for $K = 1, 2, 4, 8, 16, 32$ respectively, matching the Part 2 table. Since Part 2 concludes that $K = 32$ gives the best test performance, the most relevant single-point comparison is at $K = 32$, while the full six-curve plot is used to understand the overall pattern across model sizes.

Feature Removal at $K = 32$

At the best-performing setting $K = 32$, removing Time increases fair MAE from 9.24 to 10.25 (degradation of +1.01). Removing Temp(C) increases the MAE to 12.50 (+3.26). Removing RH(%) increases the MAE to 9.92 (+0.68). Removing o3op1(mV) increases the MAE to 13.07 (+3.83), and removing o3op2(mV) increases the MAE to 10.82 (+1.58).

Therefore, at the operating point selected from Part 2, the most important feature is o3op1(mV), followed by Temp(C), then o3op2(mV), then Time, while RH(%) has the smallest effect.

Why Does Removing Time Hurt Less?

Although removing Time degrades performance, its effect is clearly smaller than that of the ozone sensor channels and temperature. A reasonable explanation is partial redundancy: the Time feature overlaps with information already carried by variables such as Temp(C) and, to some extent, RH(%) ,

because these variables also vary systematically over the course of the day. This interpretation is supported by the feature correlations on the training split: Time has moderate positive correlation with temperature (0.4005) and moderate negative correlation with relative humidity (-0.3699). Hence, Time contributes useful information, but part of its predictive signal can already be inferred indirectly from the weather-related variables.

Features Affect Both Experts and Routing

An important machine-learning point is that feature removal affects the mixed regression model in two ways. First, it weakens the regression models learned inside each mixture component, since each expert now has less information for predicting ozone accurately. Second, it can also reduce the accuracy of the auxiliary classifier that assigns each test point to one of the learned components at test time. Therefore, the increase in fair test MAE after removing a feature reflects both poorer regression quality and poorer component assignment.

Correlation and Multicollinearity Discussion

The strong degradation caused by removing o3op1 (mV) and Temp (C) does not arise because these two features are highly collinear. In fact, their training-set correlation is only 0.1037, which is weak. Instead, they appear to contribute different but equally important information: o3op1 (mV) provides a direct ozone-related sensor signal, while Temp (C) provides important environmental context. By contrast, the strongest correlation in the feature set is between o3op1 (mV) and o3op2 (mV) (0.9343), which is expected because both are ozone sensor channels.

Discussion of the Full Six-Curve Plot

The full six-curve plot reveals several useful patterns. First, the curve obtained by removing RH(%) is slightly better than the all-features baseline at $K = 2$, which suggests that humidity can be mildly redundant or noisy for some low-capacity settings, even though it hurts performance overall at larger K . Second, removing o3op1 (mV) or o3op2 (mV) hurts much more once the model has more than one component; at $K = 1$ the degradation is negligible or even slightly negative, but for $K \geq 2$ it becomes clearly positive and often large. This indicates that the mixture model exploits ozone-sensor information much more effectively once multiple components are available. Third, the penalty for removing Time remains moderate across all values of K , which is again consistent with partial redundancy between Time and the weather-related variables.

Overall Interpretation

Looking across the full set of experiments, the overall conclusion is clear: the ozone sensor channels are the most influential features, temperature is also strongly informative, and time and relative humidity provide weaker supporting information. A concise summary of the feature-importance ranking suggested by the best-performing $K = 32$ model is

$$\text{o3op1 (mV)} > \text{Temp (C)} > \text{o3op2 (mV)} > \text{Time} > \text{RH(\%)}.$$

Thus, the mixture model relies most strongly on the ozone sensor readings and temperature for accurate prediction, while time and humidity mainly provide secondary context.

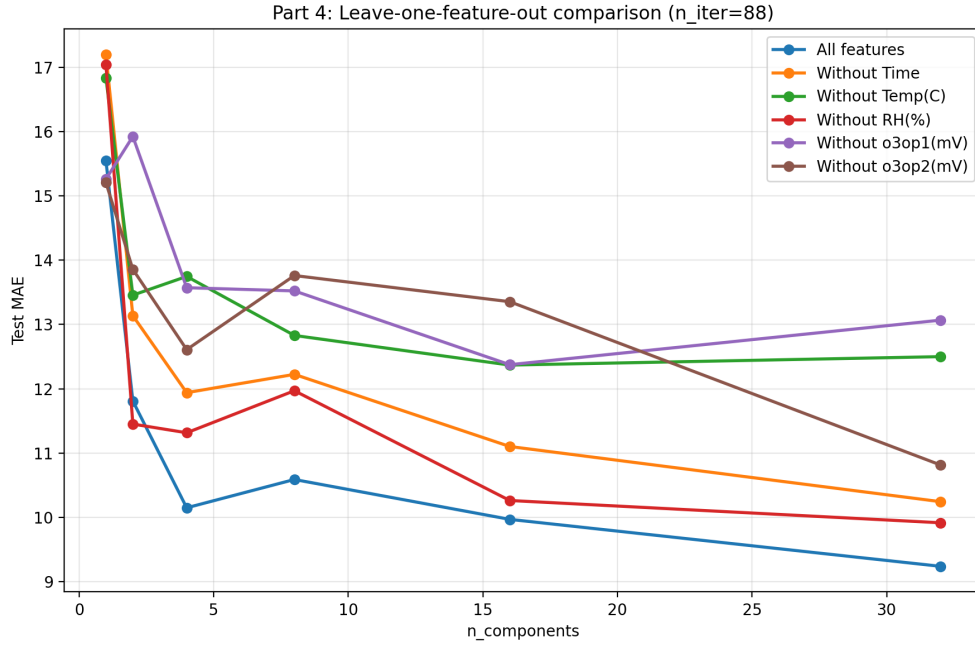


Figure 5: Fair test MAE versus number of mixture components for the all-features model and the five leave-one-feature-out variants, using $n_{\text{iter}} = 88$. Removing the ozone sensor channels and temperature causes the largest degradation, while the RH curve shows mild cross-over behavior at small K .

Part 5: Mixture vs. a Flat Decision Tree

A mixed-regression model with K components and an auxiliary classifier is similar in spirit to a shallow regression tree: the classifier routes each test point to one of K experts, and each expert makes the final prediction. The key difference is that the mixture uses *ridge-regression* experts, which can extrapolate linearly inside each regime, whereas `DecisionTreeRegressor` uses *constant-valued* leaves and learns axis-aligned splits. Part 5 tests whether a plain regression tree gives a better speed–accuracy trade-off on the AQI dataset.

5.1 Hyperparameter Tuning

We use all five features:

Time, Temp(C), RH(%), o3op1(mV), o3op2(mV).

The same 80–20 time-ordered train-test split and z -normalisation pipeline as Parts 1–4 are used.

At `max_depth=8`, we tuned the tree over

```
criterion ∈ {squared_error, friedman_mse, absolute_error, poisson},
splitter ∈ {best, random},
min_samples_split ∈ {2, 10, 50},
min_samples_leaf ∈ {1, 5, 10},
max_features ∈ {None, sqrt, log2}.
```

The `poisson` criterion is not applicable because `sklearn` requires non-negative targets for Poisson regression trees, while our z -normalised labels can be negative.

The best configuration was

```
criterion=squared_error, splitter=best, min_samples_split=2, min_samples_leaf=1, max_features=
```

Table 6: Hyperparameter search for `DecisionTreeRegressor` at `max_depth=8`. Capacity hyperparameters are fixed to their best values: `min_samples_split=2`, `min_samples_leaf=1`, `max_features=None`.

criterion	splitter	Test MAE (ppb)
squared_error	best	10.91
squared_error	random	14.86
friedman_mse	best	12.49
friedman_mse	random	14.86
absolute_error	best	11.96
absolute_error	random	13.81
poisson	best	— (signed labels)
poisson	random	— (signed labels)

5.2 Depth Sweep and Speed–Accuracy Plot

Using the best configuration, we sweep

`max_depth` ∈ {2, 3, 4, 6, 8, 10, 14, 20}.

For each depth, we record test MAE and total prediction time over the $n_{\text{test}} = 2092$ held-out points. We compare these trees with the Part 2 mixture models at `n_iter=88` for $K \in \{1, 2, 4, 8, 16, 32\}$. Figure 6 plots test MAE against prediction time on a log-scale x-axis. Lower-left is better.

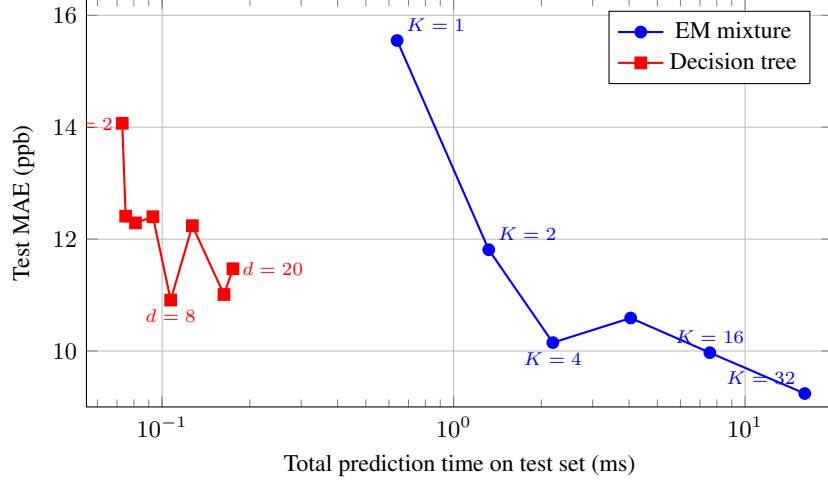


Figure 6: Speed-accuracy comparison on the full test set ($n_{\text{test}} = 2092$). Blue circles show EM mixture models for varying K at $n_{\text{iter}} = 88$, and red squares show `DecisionTreeRegressor` for varying `max_depth`. Representative points are labelled; the full depth sweep is reported in Table 7.

Table 7: `DecisionTreeRegressor` depth sweep with `criterion=squared_error` and `splitter=best`.

max_depth	Test MAE (ppb)	Predict time (ms)	# leaves
2	14.07	0.073	4
3	12.41	0.075	8
4	12.29	0.081	16
6	12.40	0.093	64
8	10.91	0.107	238
10	12.24	0.127	730
14	11.01	0.163	3176
20	11.47	0.175	6760

Table 8: Direct comparison for the tree analogy. A $K = 4$ mixture has four experts, so the closest binary decision-tree analogue is a depth-2 tree with four leaves.

Model	Effective leaves/experts	Test MAE (ppb)	Predict time (ms)
Decision tree, <code>max_depth=2</code>	4 leaves	14.07	0.073
Mixture model, $K = 4$	4 ridge experts	10.15	2.19

Table 9: Best accuracy comparison after the Part 2 update.

Model	Best setting	Test MAE (ppb)	Predict time (ms)
Decision tree	<code>max_depth=8</code>	10.91	0.107
Mixture model	$K = 32, n_{\text{iter}} = 88$	9.24	16.02

5.3 Discussion

Direct answer to the tree analogy. The assignment’s analogy suggests comparing a mixture with K experts to a tree with a similar number of terminal regions. For the natural four-expert case, a $K = 4$ mixture is closest to a depth-2 binary tree, which has four leaves. This direct comparison favours the mixture: the depth-2 tree obtains 14.07 ppb MAE, whereas the $K = 4$ mixture obtains

10.15 ppb MAE. Thus, at the same number of terminal regions, the mixture is substantially more accurate because each region contains a ridge regressor rather than a constant mean.

The updated Part 2 result selects $K = 32$ as the most accurate mixture at $n_iter = 88$. A fully comparable binary tree with exactly 32 leaves would correspond roughly to depth 5. Although depth 5 was not one of the reported sweep points, the nearby tree depths do not close the gap: depth 4 has 16 leaves and 12.29 ppb MAE, while depth 6 has 64 leaves and 12.40 ppb MAE. The $K = 32$ mixture achieves 9.24 ppb MAE, so even giving the tree a comparable or larger number of leaves does not match the mixture's accuracy.

Speed-accuracy trade-off. The decision tree is best in the ultra-fast regime. A depth-3 tree gives 12.41 ppb MAE in only 0.075 ms, while the cheapest mixture, $K = 1$, takes 0.64 ms and gives 15.55 ppb MAE. Therefore, if latency is the main constraint, the tree is the better choice.

The mixture is better when accuracy is the main constraint. The best tree found is $max_depth = 8$, with 10.91 ppb MAE at 0.107 ms. The updated best mixture, $K = 32$ at $n_iter = 88$, improves the MAE to 9.24 ppb, but takes 16.02 ms. Hence the mixture does not Pareto-dominate the tree, because it is slower; instead, it offers a better accuracy point when higher latency is acceptable.

Why does the tree saturate near 11 ppb? A regression tree is a piecewise-constant estimator: every leaf predicts only the mean response of the training points routed to that leaf. If the true relationship between the sensor readings and O_3 is locally linear, a tree can only approximate it by making many small axis-aligned boxes. This creates a bias-variance trade-off. Shallow trees have high bias because their leaves are too coarse, while very deep trees reduce training error but can overfit local noise.

The non-monotone depth curve is also explained by the tuned capacity settings. The best tree uses $min_samples_leaf = 1$, so once the depth becomes large, the tree is allowed to create very small leaves, including leaves supported by only one or a few training points. This can reduce training error but makes the leaf means unstable on the test set. That is why the MAE improves up to depth 8, worsens at depth 10, and only partially recovers at depth 14.

The mixture has a stronger leaf model. Each component is a full ridge regressor, so each routed region can still model a linear trend. This extra expressiveness explains why the mixtures beat both the four-leaf tree and the best deeper tree in accuracy.

Why is the mixture slower? A decision tree requires only $O(\text{depth})$ comparisons per test point. The mixture first uses the auxiliary classifier to route the point and then evaluates the regression expert. In the lecture-style implementation, all ridge experts are scored, so the cost is approximately $O(K + \text{classifier cost})$. This is why mixture prediction time grows from 0.64 ms at $K = 1$ to 16.02 ms at $K = 32$, while tree prediction time remains below 0.2 ms for every tested depth.

Practical recommendation. If sub-millisecond inference is required, use the depth-8 decision tree: 10.91 ppb MAE at about 0.107 ms. If prediction accuracy is more important and tens of milliseconds are acceptable, use the $K = 32$ mixture at $n_iter = 88$: 9.24 ppb MAE at about 16.02 ms. Thus, the final choice depends on the latency-versus-accuracy constraint.